■# PAPER_317: Orion M42 Trapezium Wind Ram Pressure Dominance

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η*wind = 28.47* | terosion = 467 kyr | a_wind = 5.424×10^{-10} m/s^2

FIRST UQFF HII Region Ram Pressure Dominance Ratio

Session: 91 **Module:** ORION*UQFF*MODULE.cpp (33rd C++ module) **System:** Orion Nebula M42/NGC 1976 — compact HII region, Trapezium OB cluster ionization source

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

Within the UQFF framework, this paper quantifies the ram pressure dominance of the Trapezium-driven stellar wind over self-gravity in the Orion Nebula. The dimensionless wind-gravity ratio **η_wind = P_ram/P_grav = 28.47** demonstrates that the HII region was born in a wind-dominated (unbound) state. The erosion timescale **t_erosion = 467 kyr** shows that protoplanetary discs (proplyds) currently observed at ~300 kyr age survive inside a wind-dominated environment. This is the FIRST UQFF computation of an HII region ram pressure dominance ratio.

System Parameters

Parameter	Value	Description
M	2000 M_sun = 3.978×10^{33} kg	Total nebular mass
r	1.18×10^{1}■ m (~12.5 ly)	Half-span
ρ_fluid	1×10^{-20} kg/m^3	HII gas density
v_wind	8×10^{3} m/s	Ionization front expansion speed
t_age	3×10■ yr = 9.467×10^{12} s	Nebula age
G	6.6743×10^{-11} m^3 kg^{-1} s^{-2}	Gravitational constant

Key Equations

Base Gravity

```
g_{\rm base} = \frac{GM}{r^2} = \frac{6.6743\times10^{-11} \times
3.978\times10^{33}}{(1.18\times10^{17})^2} = 1.907\times10^{-11}\ \text{m/s}^2
```

Ram Pressure Acceleration (t = 0)

```
a_{\rm wind}(t=0) = \frac{v_{\rm wind}^2}{r} = \frac{(8\times10^3)^2}{1.18\times10^{17}}
= 5.424\times10^{-10}\ \text{m/s}^2
```

Ram Pressure Acceleration (t = t_age)

$$a_{\rm wind}(t_{\rm age}) = \frac{v_{\rm wind}^2}{r}\left(1 + \frac{t_{\rm age}}{t_{\rm age}}\right) = 2 \times 5.424\times10^{-10} = 1.085\times10^{-9}\ \text{m/s}^2$$

Ram Pressure (Pa)

$$P_{\rm ram} = \rho \cdot v_{\rm wind}^2 = 10^{-20} \times (8\times10^3)^2 = 6.4\times10^{-13}\ \text{Pa}$$

Gravitational Pressure (Pa)

$$P_{\rm grav} = \frac{GM\rho}{r} = \frac{6.6743\times10^{-11} \times 3.978\times10^{33} \times 10^{-20}}{1.18\times10^{17}} = 2.248\times10^{-14}\ \text{Pa}$$

Wind–Gravity Dominance Ratio (PAPER_317 Key Result)

$$\eta_{\rm wind} = \frac{P_{\rm ram}}{P_{\rm grav}} = \frac{a_{\rm wind}}{g_{\rm base}} = \frac{5.424\times10^{-10}}{1.907\times10^{-11}} = \boxed{28.47}$$

Erosion Timescale

$$t_{\rm erosion} = \frac{r}{v_{\rm wind}} = \frac{1.18\times10^{17}}{8\times10^3} = 4.675\times10^{13}\ \text{s} = \boxed{467\ \text{kyr}}$$

Kinetic-to-Gravitational Energy Ratio

$$\frac{W_{\rm KE}}{W_{\rm grav}} \approx \frac{a_{\rm wind}}{g_{\rm base}} = 28.47$$

Results Summary

Quantity	Value	Significance
g_base	1.907×10^{-11} m/s^2	Newtonian self-gravity
a_wind(t=0)	5.424×10^{-10} m/s^2	Initial ram pressure
η_wind(t=0)	**28.47**	Wind >> gravity at birth
a*wind(*tage)	$1.085\times10^{-\blacksquare}$ m/s^2	Ram pressure at 300 kyr
η*wind(*tage)	**56.9**	Wind dominance doubles
t_erosion	**467 kyr**	Pröplyd lifetime > t_age
P_ram	6.4×10^{-13} Pa	Ram pressure
P_grav	$2.248\times10^{-1\blacksquare}$ Pa	Gravitational pressure

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UQFF Physical Interpretation

The Orion Nebula at t*age = 300 kyr is still* ***28.5–57× wind-dominated*** *(*ηwind ranging from 28.47 at t=0 to 56.9 at t*age). The erosion timescale* terosion = 467 kyr > t_age = 300 kyr confirms that proplyds currently observed by HST survive because they have not yet been fully ablated by the ram pressure — consistent with observational evidence of ~150–180 proplyds in the Orion Nebula.

UQFF Significance: This is the FIRST UQFF quantification of wind ram pressure dominance over self-gravity in a compact HII region. The wind term `a_wind(t) = v_wind²/r × (1+t/t_age)` [registered as WOLFRAM*TERM*ORION*WIND*RAM in ORION*UQFF*MODULE.cpp] dominates the gravitational term by a factor of 28.47 at birth, growing to 56.9 at 300 kyr — placing Orion in the same UQFF "wind-dominant" class as post-AGB bipolar PNe but via HII ionization physics rather than stellar wind shocks.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_WIND_RAM(val) (val)
// wind = v_wind^2/r * (1+t/t_age) [PAPER_317 ram pressure dominance; eta_wind=28.47]
```

Series first: FIRST UQFF HII region ram pressure dominance ratio. Distinguishes compact OB-driven HII (η_{wind}=28.47) from extended GMC HII and from bipolar PN wind shocks (η_{wind}~7×10■, PAPER_311).